# Blessed G/Michael Catholic Secondary School

## Semester 1 Agriculture Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

**Name:** _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

**I. Write 'True' if the statement is correct and 'False' if it is incorrect.**

____ **1.** Leguminous crops enrich soil fertility primarily by fixing atmospheric nitrogen through symbiotic Rhizobium bacteria in their root nodules.

____ **2.** Drip irrigation has the lowest water-use efficiency among all contemporary irrigation methods.

____ **3.** Soil salinization occurs frequently in arid and semi-arid regions characterized by poor drainage and excessive surface evaporation.

____ **4.** Colostrum is the first milk produced by mammals after giving birth, containing concentrated antibodies essential for neonatal passive immunity.

____ **5.** Integrated Pest Management (IPM) advocates the total eradication of all pest insects exclusively using broad-spectrum synthetic chemical pesticides.

**II. Choose the best answer among the given alternatives.**

____ **6.** Which major cereal crop is indigenous to Ethiopia and serves as the primary staple grain for making injera?

**A.** Wheat (Triticum aestivum) **B.** Teff (Eragrostis tef)

**C.** Maize (Zea mays) **D.** Barley (Hordeum vulgare)

____ **7.** What is the primary visual symptom of severe nitrogen (N) deficiency in cereal crops?

**A.** Purplish discoloration on the lower leaf margins
**B.** Interveinal chlorosis on the youngest top leaves
**C.** General yellowing (chlorosis) starting on older lower leaves along the midrib
**D.** Marginal scorching and necrotic brown leaf tips

____ **8.** Phosphorus deficiency in plants is typically characterized by:

**A.** Purplish or reddish discoloration on stems and older leaves
**B.** Bleaching of flower petals
**C.** Stunted root growth with white spots
**D.** Rapid wilting under full sunlight

____ **9.** Which soil order is prevalent in the central highlands of Ethiopia, known as heavy black clay soils that crack when dry and expand when wet?

**A.** Vertisols **B.** Nitisols **C.** Arenosols **D.** Ferralsols

____ **10.** What broadbed maker (BBM) technology was specifically developed by Ethiopian agricultural researchers to manage Vertisol soils?

**A.** To drill deep boreholes for groundwater extraction
**B.** To drain excess surface water and construct raised seedbeds for early planting
**C.** To apply liquid pesticides automatically
**D.** To crush rocks in steep mountainous terrains

____ **11.** The practice of growing two or more crops simultaneously on the same piece of land in distinct rows is known as:

**A.** Monoculture **B.** Intercropping

**C.** Shifting cultivation **D.** Fallowing

____ **12.** Which of the following describes the ruminant stomach compartment where microbial fermentation of cellulose occurs?

**A.** Abomasum (true stomach) **B.** Rumen

**C.** Omasum **D.** Reticulum alone

____ **13.** Which stomach compartment in ruminants secretes hydrochloric acid and gastric enzymes, functioning similarly to the monogastric stomach?

**A.** Rumen **B.** Reticulum **C.** Omasum **D.** Abomasum

____ **14.** Silage is preserved green forage feed prepared through which biological process under anaerobic conditions?

**A.** Lactic acid fermentation
**B.** Aerobic mold composting
**C.** Thermal sun drying
**D.** Alkaline neutralization

____ **15.** Which fungal disease represents one of the most destructive threats to wheat production in the Ethiopian highlands?

**A.** Stem rust (Puccinia graminis f. sp. tritici)
**B.** Downy mildew of onion
**C.** Coffee berry disease
**D.** Bacterial wilt of enset

____ **16.** Enset (Ensete ventricosum), commonly termed the 'false banana', is a drought-resilient multi-purpose staple crop primarily cultivated in:

**A.** The arid Somali lowlands
**B.** The Southern and Southwestern regions of Ethiopia
**C.** The Afar depression
**D.** The Gambella wetlands

____ **17.** What devastating plant disease threatens Enset cultivation and is caused by the bacterium Xanthomonas vasicola pv. musacearum?

**A.** Bacterial Wilt of Enset (Xanthomonas wilt)
**B.** Coffee Leaf Rust
**C.** Late Blight
**D.** Smut disease

____ **18.** Soil erosion caused by rainwater runoff can be effectively mitigated on steep hillslopes through:

**A.** Deep contour plowing, terracing, and vegetative grass strips
**B.** Intensive downhill tillage along the slope
**C.** Overgrazing hillsides to compact topsoil
**D.** Complete deforestation of hillside trees

____ **19.** Which of the following organic amendments is most effective for improving both water retention in sandy soils and aeration in clay soils?

**A.** Urea chemical fertilizer
**B.** Decomposed compost or farmyard manure (organic matter)
**C.** Ground limestone (calcium carbonate)
**D.** Sodium chloride crystals

____ **20.** The economic principle stating that adding successive units of a variable input (such as fertilizer) to fixed land eventually yields smaller additions to output is:

**A.** The Law of Comparative Advantage
**B.** The Law of Diminishing Marginal Returns
**C.** The Law of Supply and Demand
**D.** The Economies of Scale

____ **21.** Which artificial insemination (AI) procedure is standard in commercial dairy cattle breeding?

**A.** Vaginal natural mating only
**B.** Recto-vaginal cervical deposition using cryopreserved straw semen
**C.** Surgical embryo implantation in the abomasum
**D.** Spontaneous hormone induction without semen

____ **22.** What term denotes the removal of unthrifty, diseased, or unproductive livestock from a farm herd to enhance herd genetic quality and profitability?

**A.** Castration **B.** Culling **C.** Docking **D.** Dehorning

____ **23.** Which of the following feeds is categorized as a concentrated protein supplement for livestock rations?

**A.** Barley straw
**B.** Nougcake (Guizotia abyssinica meal)
**C.** Corn cobs
**D.** Fresh elephant grass

____ **24.** In poultry production, which viral disease causes acute respiratory distress, nervous tremors, paralysis, high mortality, and requires mandatory vaccination?

**A.** Newcastle Disease
**B.** Fowl Coccidiosis
**C.** Infectious Coryza
**D.** Scaly Leg Mites

____ **25.** Which agroforestry system combines woody perennial trees with agricultural field crops and pasture animals on the same management unit?

**A.** Silvopasture / Agrosilvopastoral system
**B.** Monoculture forestry
**C.** Hydroponics
**D.** Open-cast mining

____ **26.** What is the primary objective of post-harvest hermetic storage bags (such as Purdue Improved Crop Storage - PICS bags)?

**A.** To accelerate seed germination in storage
**B.** To suffocate storage insect pests by depleting oxygen and trapping $CO_2$ without chemical fumigants
**C.** To bleach grains to improve appearance
**D.** To raise storage humidity to encourage fermentation

____ **27.** In farm accounting, what document records all assets, liabilities, and owners' equity of an agricultural enterprise at a specific point in time?

**A.** Balance Sheet
**B.** Income (Profit & Loss) Statement
**C.** Cash Flow Statement
**D.** Physical Inventory Log

____ **28.** A parasitic weed that attaches to the roots of sorghum and maize, causing severe chlorosis and yield loss in sub-Saharan Africa, is:

**A.** Striga hermonthica (Witchweed)
**B.** Parthenium hysterophorus
**C.** Lantana camara
**D.** Eichhornia crassipes

____ **29.** What is the primary purpose of subsoiling or deep ripping in agricultural soil management?

**A.** To bury crop seeds under 50 cm of soil
**B.** To fracture dense compacted plow pans and enhance water percolation and root penetration
**C.** To kill earthworms and microorganisms
**D.** To bring bedrock stones to the soil surface

____ **30.** Which honeybee caste develops from unfertilized eggs through arrhenotokous parthenogenesis?

**A.** Drone (male bee)
**B.** Worker bee (sterile female)
**C.** Queen bee (fertile female)
**D.** Royal nurse bee